

Algobetting Strategy

A reproducible quantitative research platform for testing whether apparent value in sports betting markets survives margin removal, calibration, uncertainty, execution constraints, and genuinely out-of-time validation.

MATHEMATICS

De-vigging, consensus, calibration, EV, Kelly

ENGINEERING

Immutable data, normalization, provenance

VALIDATION

Point-in-time replay, holdouts, risk gates

Prepared 8 August 2026 | Repository v0.1.0 | Evidence snapshot 567a958
Research software only. No live bet placement. No guarantee of profitability.



Executive investment snapshot

The project is investable as research infrastructure, not as a proven betting fund.

Core proposition. Bookmaker prices contain both information and margin. Algobetting Strategy turns point-in-time odds into auditable probability estimates, then tries to falsify apparent edges with calibration, uncertainty bounds, closing-price comparison, and realistic staking constraints.

2	146	13,101	0
SPORT-SPECIFIC PIPELINES	AUTOMATED TESTS PASS	PYTHON LINES IN SRC + TESTS	LIVE BETTING ENDPOINTS

What exists now

- A shared two-outcome pricing core with point-in-time market selection, proportional and power de-vigging, median consensus, fair odds, and expected value.
- A tennis application with provider ingestion, conservative entity resolution, immutable SQLite history, point-in-time signal evaluation, and a completed 127-match Wimbledon research export.
- An NBA moneyline pipeline with exact cost preflight, resumable raw-data cache, chronological calibration, weekly block bootstrap uncertainty, closing value, detailed CSV/JSON reporting, and locked acceptance gates.
- A clean research boundary: the system does not place bets and does not represent apparent model edge as guaranteed profit.

Investor interpretation

Asset	Current evidence	Commercial implication
Research engine	Working, tested, sport-separated architecture	Reusable foundation for additional two-outcome markets
Tennis evidence	Completed but strongly negative Wimbledon baseline	Shows falsification discipline; strategy is not yet investable capital
NBA evidence	Rigorous pipeline and preflight; paid odds pending	Funding accelerates the next decisive experiment
Defensibility	Provenance, normalization, point-in-time replay, explicit gates	Trustworthy research process is the primary moat today

The problem being solved

Sports markets make probability look simple: take the reciprocal of decimal odds. In reality, the quoted prices embed margin, providers disagree, timestamps matter, markets can be stale or suspended, and a backtest can look profitable merely because it used information that was not available when a decision would have been made.

The information gap

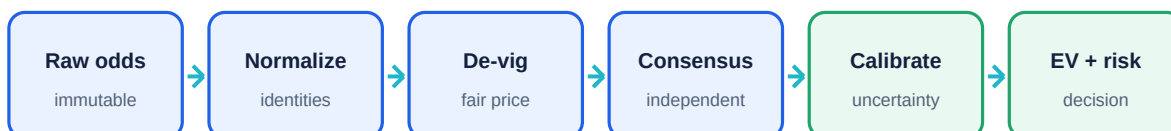
Market failure	Naive response	Algo betting response
Bookmaker margin	Treat 1/O as fair probability	Remove overround before comparing prices
Self-referential consensus	Let the target book influence its benchmark	Leave the evaluated bookmaker out
Stale or future data	Use the latest row in the database	Reconstruct the latest knowable state at decision time
Noisy probability	Stake from a point estimate	Calibrate, bootstrap, and use a lower probability bound
Backtest optimism	Optimize thresholds on the same sample	Chronological splits, untouched holdouts, fixed acceptance gates
Execution gap	Assume every displayed quote is tradable	Roadmap includes slippage, limits, rejection, latency, and open exposure

The commercial opportunity is therefore broader than finding a bet. The same machinery can produce research-grade pricing diagnostics, data-quality audits, model-validation reports, and transparent educational tools.

Central hypothesis, not promise: under tightly specified conditions, can an independently constructed and uncertainty-adjusted probability estimate identify offers whose advantage survives later market information, execution frictions, and an untouched time period?

Product: a research operating system for market odds

POINT-IN-TIME RESEARCH PIPELINE



Every derived record retains source, snapshot time, method, parameters, and calculation time.

One engine, two applications

Layer	Tennis	NBA	Shared principle
Raw data	The Odds API and OddsPapi adapters	BALLDONTLIE schedules + The Odds API history	Preserve exact provider responses
Identity	Players, tournaments, bookmakers, matches	Canonical NBA teams and games	Fail closed on ambiguity
Entry	Live research or T-60 Wimbledon replay	T-60 before tip	Only information available at decision time
Benchmark	Pinnacle sharp proxy or leave-one-out median	Leave-one-out median	Do not let a quote define itself
Risk	One bet per match, open-cash reservation, 25% Kelly	p_safe, one candidate per game, 25% Kelly sensitivity	Probability quality precedes sizing
Output	CLI + two detailed CSV exports	Seven CSV/JSON report layers	Manual inspection remains possible

Who can use the platform

B2B	B2C	API	LAB
MODEL & DATA QA	RESEARCH EDUCATION	PRICING DIAGNOSTICS	REPRODUCIBLE EXPERIMENTS

The near-term product is a research and validation layer. Any later wagering or execution product would require separate technical controls, capital policy, compliance review, and jurisdiction-specific legal advice.

Mathematical core I: from odds to fair probabilities

1. Implied probability

For decimal odds O_i , the raw implied probability is the reciprocal:

$$q_i = 1 / O_i$$

2. Overround

For a complete two-outcome market, the quoted probability mass is:

$$R = q_1 + q_2$$

R above 1 is the embedded book margin under this simplified interpretation. The project keeps the observed overround as an auditable quality signal rather than hiding it.

3. Proportional de-vigging

The transparent baseline normalizes the implied probabilities:

$$p_i = q_i / (q_1 + q_2)$$

4. Power de-vigging

The active implementations can solve for a common exponent k so that the transformed probabilities sum to one:

$$\text{find } k \text{ such that } q_1^k + q_2^k = 1; \text{ then } p_i = q_i^k$$

Why the power method matters. It can redistribute margin non-linearly across favourites and underdogs. The code solves k deterministically with bounded bisection and Decimal arithmetic, then renormalizes the final pair to sum exactly to one.

Worked example

Input	Calculation	Interpretation
Home 1.80	$q_{\text{home}} = 55.56\%$	Raw implied probability
Away 2.10	$q_{\text{away}} = 47.62\%$	Raw implied probability
Total	$R = 103.17\%$	3.17 percentage points above 100%
Proportional fair	53.85% / 46.15%	A transparent de-vigged baseline

Mathematical core II: consensus, fair odds, EV, and closing value

Independent consensus

For bookmaker b and outcome i , the NBA baseline takes the median of every other eligible bookmaker's de-vigged probability. This prevents b from mechanically pulling its own comparison benchmark toward its quote.

$$p_{\text{consensus}}(i, b) = \text{median}(\{p_{i,j} : j \neq b\})$$

Tennis also supports a designated sharp-book proxy. In the active research profile, Pinnacle is power-de-vigged and used as the reference for other books; Pinnacle is never evaluated against itself. This is a proxy, not ground truth.

Fair odds and expected value

$$\text{fair_odds} = 1 / p_{\text{consensus}}$$

$$\text{EV} = o_{\text{offered}} * p_{\text{consensus}} - 1$$

EV is expected profit per unit staked under the model probability. An EV threshold is a research filter, not proof that the offer is executable or that the probability is calibrated.

Closing value

$$\text{CLV} = o_{\text{entry}} * p_{\text{close}} - 1$$

Positive closing value means the entry odds look favorable relative to the later de-vigged market estimate. It is a useful process diagnostic because it does not require a single match outcome to be lucky, but it remains only a proxy for pricing quality.

Decision hierarchy: a positive raw EV is only the start. The investable research claim requires calibrated probabilities, a conservative uncertainty bound, executable odds, enough sample size, positive closing value, and positive holdout performance with a positive lower confidence bound.

Calibration and uncertainty: making EV harder to earn

Identity-anchored logistic calibration

The NBA pilot fits one independent home-win observation per game. It maps the market probability through a regularized logistic model:

$$\text{logit}(p_{\text{cal}}) = a + b * \text{logit}(p_{\text{market}})$$

The penalty shrinks a toward 0 and b toward 1, the identity mapping. This is deliberately conservative: limited data should not let the calibration layer invent a large correction.

Chronological separation

Phase	Share	What it can do	What it cannot do
Training	60%	Fit calibration for validation	Generate betting candidates
Validation	20%	Test first calibrator; join training for test fit	Change the locked test threshold
Test	20%	Evaluate the refit model once	Feed information back into earlier phases

Weekly block-bootstrap lower bound

Two hundred deterministic bootstrap refits resample seven-day chronological blocks. For the selected side, the decision probability is capped at the lower fifth percentile of the bootstrap distribution:

$$p_{\text{safe}} = \min(p_{\text{cal}}, \text{quantile_5\%}(\{p_{\text{bootstrap}}, m\}))$$

$$EV_{\text{safe}} = O_{\text{executable}} * p_{\text{safe}} - 1$$

The opposite side is always derived as $1 - p_{\text{home}}$. It is not calibrated independently. That preserves the logical constraint that the two outcomes sum to one and avoids double-counting each game.

Scoring rules

$$\text{Brier} = \text{mean}((p_t - y_t)^2) \mid \text{Log loss} = -\text{mean}(y_t \log p_t + (1 - y_t) \log(1 - p_t))$$

Staking and portfolio risk

Kelly sizing

For net decimal profit $b = O - 1$, win probability p , and loss probability $1 - p$, full Kelly is:

$$f^* = (b \cdot p - (1 - p)) / b = (O \cdot p - 1) / (O - 1)$$

The current baseline applies one quarter of full Kelly:

$$\text{stake} = \text{available_equity} * 0.25 * \max(0, f^*)$$

Risk control	Tennis status	NBA status	Next institutional control
One position per event	Implemented	Implemented	Portfolio-wide correlation clusters
Open cash reservation	Implemented using actual settlement timestamps	Not yet event-driven	Placement/settlement ledger with reserved capital
Conservative probability	Not calibrated in current tennis strategy	Implemented in calibrated pilot	Unified p_{safe} contract across sports
Stake fraction	25% Kelly	25% Kelly sensitivity	Per-position, daily, bookmaker, and drawdown caps
Execution costs	Not modeled	Not modeled	Slippage, rejection, commission, latency, limits

Critical principle. Fractional Kelly reduces volatility only when p is approximately right. It does not repair a biased or uncalibrated probability. Calibration and uncertainty therefore sit before staking in the product architecture.

Target institutional formula

$$f_{\text{position}} = \min(f_{\text{cap}}, k * \max(0, (O_{\text{exec}} \cdot p_{\text{safe}} - 1) / (O_{\text{exec}} - 1)))$$

This remains subject to reserve capital, open exposure, concentration, correlation haircuts, bookmaker limits, and a drawdown policy. Those controls are a roadmap requirement, not a claim about the current NBA settlement implementation.

Data integrity: the real foundation of the edge

Point-in-time rule

At decision time t , the system may select only a quote observed at or before t . It selects the latest state first; if that state is inactive, invalid, or stale, the market is unavailable. It never resurrects an older active price after a later suspension.

Provenance chain

Control	Implementation behavior	Why investors should care
Raw preservation	Exact bytes or JSON cached before transformation	Replay and vendor dispute audit
Stable identity	Deterministic provider/event and snapshot identifiers	Idempotence and duplicate control
Conservative resolution	Unknown or ambiguous player/team names are quarantined	Avoid silent entity merges
UTC timestamps	Timezone-aware UTC for persistence and decisions	Comparable chronology across vendors
Observable exclusions	Future, stale, incomplete, duplicate, and conflicting quotes are recorded	No invisible denominator manipulation
Atomic cache	Successful responses and explicit no-data states can be resumed	Controlled cost and failure recovery
Paid-data gate	Exact quota preflight and matching confirmation required	Prevents accidental spend

Quality flags

EDGE UNUSUALLY LARGE EV	MARGIN SUSPICIOUS OVERROUND	RANGE WIDE PEER DISPERSION	CONFLICT SAME-TIME PRICE DISAGREEMENT
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Flags remain attached to research rows. The design prefers an auditable anomaly to a silently deleted observation, while market-level contradictions fail closed.

Tennis application: current operating state

2 DATA PROVIDERS	9 WIMBLEDON BOOKMAKERS	T-60 FIXED DECISION TIME	25% KELLY FRACTION
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Implemented capabilities

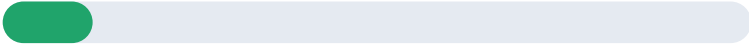
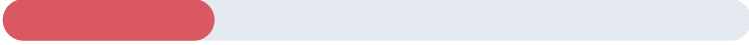
- The Odds API collection for current match-winner markets, raw response preservation, explicit player approval, normalization, SQLite history, and point-in-time evaluation.
- OddsPapi historical Wimbledon workflow with resumable fixture, settlement, and odds caches; three groups of three bookmakers per historical request; scheduled start rather than true start for T-60 decisions.
- Power de-vigging; active Pinnacle sharp-reference mode; an alternative leave-one-out median mode; all eligible offers returned with flags and provenance.
- One selected position per match, void/missing-result handling, 25% Kelly simulation, and capital reservation until actual settlement time.
- Two inspection-ready exports: one match row per fixture and one offer row per player-bookmaker pair.

Known research limitations

Limitation	Effect	Required response
Pinnacle reference is not ground truth	A delayed or stale sharp feed can create false divergence	Measure vendor delay and compare alternative consensus methods
Historical replay has no quote-age cutoff	Very old but point-in-time quotes can remain eligible	Add historical availability/active-state heartbeat or explicit age policy
No probability calibration	EV and Kelly can be overconfident	Chronological calibration and p_safe before candidate selection
One tournament sample	No season, surface, or regime generalization	Multi-season ATP/WTa walk-forward evidence
No execution model	Displayed price may not be obtainable	Log executable odds, latency, limits, rejection, and slippage

Wimbledon 2026 evidence: a useful rejection, not a success story

127 FIXTURES	2,280 OFFERS EVALUATED	59 CANDIDATE OFFERS	32 SELECTED BETS
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Flat-stake ROI		-67.2%
Kelly yield		-78.0%
Win rate		3/32
Equity change		-2,209

Evidence	Observed value	Reading
Coverage	8-9 eligible books per match	Broad bookmaker coverage in the export
Result coverage	124/127 completed	Three non-completed/other settlement states
Selected outcome profile	Odds 3.00-36.00; median 7.17	Strategy concentrated on long-priced selections
Selected EV	Median 12.7%; max 2162.7%	Maximum is economically implausible without anomaly review
Flat result	-21.49 units; ROI -67.2%	Strongly negative baseline
Quarter-Kelly	Profit -2,209; final equity 7,791	Position sizing amplified bad probability inputs
Max drawdown	2,422	Material capital-path risk

Investor conclusion. This run is not evidence of a profitable tennis strategy. Its extreme apparent edges and poor realized performance justify a formal anomaly audit covering quote age, provider latency, suspension state, outlier prices, market definition, and sharp-reference validity before any further staking claim.

NBA application: the next decisive experiment

2023-24 BOUNDED PILOT	1,484 ENTRY + CLOSE TIMESTAMPS	14,840 MAX ADDITIONAL CREDITS	0 HISTORICAL REQUESTS CACHED
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Locked pilot design

Decision	Locked setting	Purpose
Market	NBA regular-season moneyline including overtime	Single coherent two-outcome market
Entry / close	60 minutes / 5 minutes before tip	Separate decision and closing diagnostics
Freshness	Maximum quote age 30 minutes	Reduce stale-price artifacts
Coverage	At least five complete fixed-odds books	Independent market depth
Pricing	Power de-vig + leave-one-out median	Transparent robust baseline
Calibration	60/20/20 chronological split; L2 identity anchor	Prevent same-period leakage and overfit
Uncertainty	200 weekly-block bootstraps; 5th percentile	Use p_safe for EV and Kelly
Selection	5% EV_safe; at most one candidate per game	Fixed decision rule
Acceptance	Coverage + sample + CLV + ROI + lower CI gate	Avoid narrative cherry-picking

Current gate

The manifest contains 1,484 pending requests and 0 cached historical requests. The pipeline is ready for a fresh no-spend preflight, but the current repository does not contain NBA historical-odds results. Performance must therefore remain **unclaimed**.

Outputs after a completed offline backtest

- Games, offers, candidates, equity curve, exclusions, summary CSV, and summary JSON.
- Coverage, Brier score, log loss, calibration bins, closing value, flat ROI, turnover, hit rate, drawdown, confidence intervals, and Kelly sensitivity.
- Breakdowns by season, bookmaker, favourite/underdog, odds band, and EV band.

Validation framework: how a claim earns the word supported

Evidence ladder

Level	Question	Metric / gate	Current project status
1. Integrity	Was the data knowable and correctly matched?	UTC chronology, raw provenance, exclusions, match rate	Strong engineering controls
2. Pricing	Does de-vigging and consensus beat simpler baselines?	Out-of-time log loss, Brier, calibration	Baseline implemented; challenger study pending
3. Stability	Does the edge persist across time and segments?	Walk-forward blocks, seasons, books, odds bands	NBA design locked; evidence pending
4. Market confirmation	Does later market information agree?	Positive mean CLV with full coverage	Implemented as an output
5. Economic value	Does profit survive uncertainty?	Positive ROI and positive lower 95% mean-profit bound	Hard gate in five-season NBA profile
6. Execution	Can the price and stake be obtained?	Slippage, rejection, limits, latency, commission	Roadmap
7. Portfolio	Can risk be held safely?	Open exposure, concentration, drawdown, ruin stress	Partial; NBA event ledger pending

Five-season confirmatory rule

The five-season NBA profile reserves 2025-26 as holdout and requires at least 300 candidates, positive mean closing value, positive flat-stake ROI, and a positive lower 95% confidence bound, alongside complete timestamp collection and coverage thresholds. Otherwise the conclusion is negative or inconclusive.

Why this matters commercially

- A negative result is still a valid research output because the rules were locked before the result.
- Inconclusive is distinct from negative: missing data or an underpowered sample cannot be marketed as failure or success.
- Threshold changes belong in development only; the untouched holdout cannot be re-optimized after inspection.
- The model must be evaluated on all eligible games for calibration, not only offers that passed an EV threshold.

Technology architecture and engineering moat

53 PYTHON FILES IN SRC + TESTS	3 PACKAGES	19 COMMITTS IN CURRENT HISTORY	3/3 TEST + LINT + TYPE CHECK PASS
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Package boundaries

Package	Responsibility	Defensible design choice
betting_core	Sport-neutral two-way quote selection, de-vigging, median, fair odds, EV	One audited mathematical core
tennis_value	Provider ingestion, entity resolution, SQLite history, pricing, signals, replay, exports	Conservative identity and immutable odds state
basketball_value	NBA schedule/odds acquisition, normalization, calibration, strategy, reporting	Chronological uncertainty-aware research
tests	Unit, integration, end-to-end, storage, provider, calibration, reporting checks	Synthetic/recorded fixtures; no live API dependence

Quality posture

- Python 3.12, public type hints, frozen domain records, structured exceptions, Decimal odds calculations, pathlib paths, and UTC persistence.
- Ruff formatting/lint discipline, strict mypy, deterministic pytest fixtures, and regression tests for point-in-time and identity behavior.
- Secrets stay in environment variables or ignored local files; generated data, databases, caches, and reports are not committed.
- Provider-specific logic is isolated so pricing and validation can be replayed independently of acquisition.

Verification performed for this dossier at commit **567a958**: 146 tests passed; Ruff reported all checks passed; strict mypy reported no issues across 53 checked source files.

Commercial model: monetize trust before risk capital

Recommended sequencing

Stage	Customer	Offer	Revenue logic	Prerequisite
1. Research lab	Quant learners, analysts, creators	Transparent reports, explainers, reproducible synthetic examples	Membership, course, sponsorship, grant	Clear methodology and disclosure
2. Data QA	Odds/data vendors, affiliates, research teams	Timestamp, identity, stale-state, outlier, and coverage diagnostics	Subscription or per-audit fee	Vendor-compatible connectors
3. Analytics API	Sports analytics products	De-vigged consensus, anomaly scores, calibration diagnostics	Usage-based API / license	Service reliability and data rights
4. Decision support	Sophisticated research users	Watchlists, risk-adjusted signals, execution logging	Professional subscription	Validated p_safe and compliance review
5. Capital strategy	Separate regulated/contractual structure	Model-driven capital allocation	Performance economics	Multi-season evidence, execution, governance

Why this order is rational

- Research and data-quality products can create value even when a betting strategy is negative.
- They build user trust, provider knowledge, distribution, and real-world execution telemetry before risking capital.
- Commercial odds data may have redistribution limits; code, methodology, synthetic examples, and aggregate results can remain publishable where contracts permit.
- Any monetization involving personalized betting recommendations or execution should undergo jurisdiction-specific legal review before launch.

Positioning: Algotbetting Strategy is a transparent quantitative lab that helps users understand what a quoted edge is made of - and how often it disappears under better data and stricter validation.

12-month research and product roadmap

Window	Build	Evidence milestone	Go / no-go decision
0-3 months	Complete NBA pilot acquisition; event-driven exposure ledger; tennis anomaly audit	Reproducible pilot report + root-cause register	Is data quality sufficient for multi-season spend?
3-6 months	Five-season NBA collection; de-vig and consensus challengers; execution telemetry schema	Development + validation comparison locked before holdout	Does calibration and CLV improve out of time?
6-9 months	Untouched NBA holdout; multi-season tennis data; unified p_safe interface	Acceptance-gate decision; negative and inconclusive allowed	Is any strategy evidence strong enough to continue?
9-12 months	Research dashboard/API, synthetic public dataset, documentation, compliance assessment	Pilot users and reproducibility audit	Commercialize QA/analytics, pause strategy, or expand

Model challenger order

- Alternative de-vigging: proportional, power, additive/odds-ratio, and Shin-style challengers where appropriate.
- Consensus: median, trimmed mean, robust location, and reliability/freshness-weighted methods only after independent-source analysis.
- Market-residual model: $\text{logit}(p_{\text{final}}) = \text{logit}(p_{\text{market}}) + \delta(x)$, with strong shrinkage of δ toward zero.
- Sport features only when timestamped: NBA efficiency, Elo, rest, travel, injuries/lineups; tennis surface Elo/Glicko, serve/return, workload, travel, altitude, event level, injury/retirement risk.
- Closing-edge survival model as a separate label from win probability; train on all valid offers, not just selected bets.

Non-negotiable research controls

TIME	LOCK	ALL	RAW
CHRONOLOGICAL SPLITS	PREDECLARED GATES	FIT ON ELIGIBLE GAMES	PRESERVE PROVENANCE

The roadmap does not start with neural networks or large feature sets. The market remains the anchor until simple, regularized challengers beat it out of time on probability quality and closing value.

Funding deployment and investor milestones

No absolute funding ask has been supplied, so this dossier proposes a percentage-based deployment that can scale to the chosen envelope. Capital should buy decisive evidence and reusable infrastructure, not merely a larger unvalidated bet sample.

Allocation	Share	Use	Milestone unlocked
Historical data and vendor experiments	30%	Bounded NBA and tennis point-in-time history, coverage verification, contract review	Multi-season reproducible dataset
Engineering and data quality	30%	Exposure ledger, provider adapters, schema/versioning, anomaly tooling	Reliable replay and audit layer
Quantitative validation	20%	Calibration, bootstrap, challengers, walk-forward evaluation, stress tests	Locked acceptance decision
Product and distribution	10%	Research dashboard/API prototype, public methodology, pilot onboarding	External usage evidence
Compliance, security, contingency	10%	Data rights, jurisdiction review, secrets/security, vendor variance	Commercial launch readiness

Milestone-based release

Tranche	Release condition	Board/investor evidence
A	Fresh no-spend preflight and vendor coverage confirmed	Exact cost, timestamps, rights, and abort gates
B	Pilot cache complete and integrity checks pass	Coverage and exclusion report; no strategy claim yet
C	Chronological pilot published internally	Calibration, uncertainty, CLV, ROI, drawdown, limitations
D	Multi-season holdout plan locked	Frozen code/config hash and acceptance criteria
E	Commercial pilot begins	Data rights and compliance review; user value independent of profit

Suggested governance covenant: no claim of durable edge, no live capital scale-up, and no threshold change on an untouched holdout without a documented research reset approved before inspection.

Risk register and mitigations

Risk	Severity	Evidence today	Mitigation / kill criterion
False edge from stale or bad quotes	Critical	Extreme Wimbledon EVs and negative realization	Age/active-state audit; influential-book removal; anomaly quarantine
Probability miscalibration	Critical	Tennis uncalibrated; NBA pilot calibrated	Out-of-time calibration; p_safe; no Kelly from raw p
Look-ahead leakage	Critical	Point-in-time controls implemented	Timestamp lineage, chronological tests, closing data excluded from entry
Underpowered sample	High	127 tennis matches; NBA data pending	Minimum candidate gates; multi-season holdout; inconclusive allowed
Execution slippage and limits	High	Not yet modeled	Executable-odds logging, limits, latency, rejection, commission
Portfolio timing/correlation	High	Tennis cash reservation; NBA loop not event-driven	Unified exposure ledger, caps, reserve, scenario stress
Vendor/data rights	High	Commercial APIs and local raw cache	Contract review; publish code/synthetic/aggregates where permitted
Regulatory positioning	High	Research-only software, no execution	Jurisdiction counsel before advice, subscriptions, or execution
Model drift	Medium	Time-varying sports markets	Walk-forward monitoring and automatic no-go thresholds
Key-person concentration	Medium	Early-stage codebase	Documentation, tests, reproducible environments, review process

The principal business risk is not that every experiment loses. It is that a weak experiment is mistaken for evidence and scaled. The project's strongest current asset is its ability to say 'negative' or 'inconclusive' with an audit trail.

Explicit non-claims

- No guaranteed profit or claim that the house edge has been eliminated.
- No current claim of validated tennis or NBA profitability.
- No claim that Pinnacle, consensus, or closing prices equal true probability.
- No claim that quarter-Kelly makes an uncalibrated model safe.
- No live execution, personalized financial advice, or jurisdictional compliance conclusion.

Investment thesis and decision frame

Why invest

- The project already contains real engineering, not a slide-only concept: separate applications, shared mathematics, normalized domains, immutable storage, replay, calibration code, reports, and 146 passing tests.
- The research process is designed to expose false edge. The negative Wimbledon result is disclosed and converted into an explicit audit agenda.
- The next funded milestone is decisive: complete the bounded NBA experiment, then expand only if data integrity and calibration justify it.
- Commercial value can begin with analytics, data quality, education, and reproducible research before any capital strategy is credible.
- The architecture can extend to additional two-outcome sports without mixing sport-specific identity, timing, and feature assumptions.

Why wait

- There is no validated profitable strategy and no paying-customer evidence in the repository.
- Historical data costs and rights are external dependencies.
- Execution modeling, portfolio exposure, and compliance work are incomplete.
- The NBA evidence milestone is not yet complete; the strongest thesis today is research infrastructure and disciplined validation.

Recommended investor posture: fund a milestone-based research and product validation round. Underwrite the quality of the experiment, the reusable infrastructure, and the analytics pathways - not an assumed betting return.

The question for the next round

Can the platform convert disciplined data handling and conservative probability research into a repeatable analytics product - and only then demonstrate an edge that survives calibration, closing markets, execution, and time?

This document is for discussion and research review only. It is not an offer of securities, betting advice, a promise of returns, or a legal opinion.

Appendix: implementation inventory

Capability	Primary implementation	Status
Two-way market state selection	src/betting_core/two_way.py	Implemented + tested
Proportional and power de-vig	src/betting_core/two_way.py; src/tennis_value/pricing.py	Implemented + tested
Leave-one-out and sharp consensus	src/tennis_value/consensus.py; src/basketball_value/strategy.py	Implemented + tested
Tennis signals and exclusions	src/tennis_value/signals.py	Implemented + tested
Tennis persistence and identity	src/tennis_value/data/	Implemented + tested
Tennis replay and exports	src/tennis_value/oddspapi_backtest.py	Implemented; one tournament exported
Tennis open-cash Kelly	src/tennis_value/backtesting.py	Implemented + tested
NBA provider/cache workflow	src/basketball_value/providers.py; cache.py; workflow.py	Implemented + tested; odds pending
NBA logistic calibration	src/basketball_value/calibration.py	Implemented + tested
NBA bootstrap p_safe	src/basketball_value/calibration.py; strategy.py	Implemented + tested
NBA report/acceptance gate	src/basketball_value/reporting.py	Implemented + tested; evidence pending
Executable-odds model	Roadmap	Planned
Unified event-driven portfolio ledger	Roadmap	Tennis partial; NBA planned
Market-residual sports model	Roadmap	Planned
Dashboard/API	Roadmap	Planned

Repository evidence used

- Repository commit 567a958; project version 0.1.0.
- README.md and four TOML research profiles.
- Python source under src/betting_core, src/tennis_value, and src/basketball_value.
- Automated tests under tests/; 146 passed during dossier production.
- Wimbledon match and offer CSV exports under reports/wimbledon_atp_2026.
- NBA quota manifest under data/nba_moneyline; no paid historical odds cached at the evidence date.

Appendix: formula and metric glossary

Term	Formula / definition	What it answers
Implied probability	$q = 1/O$	What probability does the raw price imply?
Overround	$R = \sum(q_i)$	How much quoted probability mass exceeds 100%?
Proportional de-vig	$p_i = q_i / R$	What is a simple margin-free probability pair?
Power de-vig	choose k : $\sum(q_i^k)=1$	How can margin be redistributed non-linearly?
Consensus	median of independent de-vigged peers	Where does the robust peer market center sit?
Fair odds	$1/p$	What decimal price corresponds to probability p ?
Expected value	$O \cdot p - 1$	Expected unit profit under p
Safe expected value	$O_{\text{exec}} \cdot p_{\text{safe}} - 1$	Does edge survive uncertainty and executable price?
Closing value	$O_{\text{entry}} \cdot p_{\text{close}} - 1$	Did the entry beat the later market estimate?
Brier score	$\text{mean}((p-y)^2)$	How close are probabilities to outcomes?
Log loss	$-\text{mean}(y \log p + (1-y) \log(1-p))$	How costly is probabilistic overconfidence?
Kelly fraction	$(O \cdot p - 1) / (O - 1)$	What fraction maximizes idealized long-run log growth?
ROI / yield	profit / turnover	What was profit per unit actually staked?
Drawdown	peak equity - current equity	How far did capital fall from its prior peak?
95% mean-profit CI	$\text{mean} \pm 1.96 \cdot s / \sqrt{n}$	Is average unit profit distinguishable from zero under a normal approximation?

All equations are decision tools built on assumptions. None makes a price, probability, or return certain. The system's purpose is to expose those assumptions and test them in chronological data.

Document control

Field	Value
Evidence date	8 August 2026
Repository commit	567a958
Project version	0.1.0
Test status	146 passed; Ruff clean; strict mypy clean
Performance claim	None - Wimbledon negative; NBA odds/results pending
Prepared as	Investor and partner research dossier